Vojtěch Špaček

Contact: Brno, Czech Republic | sparesparrow@protonmail.ch | +420 605 963 739 | Book a call: calendly.com/dallheimal
Links: github.com/sparesparrow | linkedin.com/in/sparesparrow | sparesparrow.github.io/sparrow-ai-tech | citadela-resort.cz
Also: x.com/spare23sparrow | YouTube: AI-narrated audiobook of Mises' Human Action (Czech)

Senior C++ & Python Engineer · AI/LLM Integration (MCP) · Safety-Critical Embedded & HIL Verification

Pragmatic Programmer, Creative Problem Solver · Cybersecurity & Linux specialist

Profile

Senior C++ and Python engineer working on embedded, real-time and safety-critical systems: **aircraft onboard diagnostics** at Honeywell, **electron-microscopy detector software** at Thermo Fisher Scientific, a **TLS-inspecting proxy and firewall** at Trusted Network Solutions and, most recently, sole ownership of the **hardware-in-the-loop verification suite and automated bench** for a safety-certified gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54). I bring **agentic-AI tooling built on the Model Context Protocol** into regulated engineering workflows, held to the same evidence and traceability standard as the rest of the work.

Preferred Roles

- **AI/LLM Developer & Integration Engineer** — agentic systems, Model Context Protocol, prompt engineering, LLM integration.
- **Test Automation / Verification Engineer** — hardware-in-the-loop, safety-critical verification, instrument control, traceability.
- **Software Developer** — Linux, embedded, RTOS, TCP/IP, cybersecurity, high availability, HAL & IPC, backend.

AI/LLM Engineering — Approach

C++ remains my core language, but I expect AI to transform the programmer's workload, so I have deliberately grown into an AI developer: high-level languages (Python, JavaScript/TypeScript) and advanced prompt engineering. I work hands-on with all major LLM provider APIs and open-source frameworks and SDKs, integrating them into traditional tooling, applications and CI/CD with a focus on scalability, modularity, cost optimisation, reuse and automation, and I take particular care over the natural-language instructions and system prompts that get the best results from LLM agents.

Techniques: RAG, zero- and few-shot prompting, prompt caching, structured outputs, tool calling, encapsulated agents, multimodal routing and collaborative multi-agent workflows with a human in the loop.

Availability

Engagement	Independent contractor (Czech IČO, B2B) · Brno on-site, hybrid or remote · open to relocation
Work permit	EU citizen (Czech Republic): none needed in the EU/EEA · eligible for a Swiss B permit
Languages	Czech *(native)*, Slovak *(native)*, English *(C1)*, German *(A1)*

Core Skills

Programming	**C++17/20**, **Python** *(senior)* · **C#**, **JavaScript/TypeScript** *(medior)* · **Rust**, **Qt/QML** *(working knowledge)*
Embedded	ARM, RTOS, state machines, HAL & IPC (COM/MIDL, Unix sockets), FPGA, gRPC, FlatBuffers
Verification & HIL	HIL benches, `pyvisa/USBTMC/SCPI`, `CAN`, `Modbus`, traceability, UL/CSA test evidence
Linux & security	`Netlink`, `nftables`, TCP/IP, TLS 1.3 inspection, Kerberos/LDAP/OCSP, Suricata, PostgreSQL
AI / LLM	MCP servers, agentic workflows, tool calling, RAG, prompt engineering, LLM provider APIs
Build & delivery	Conan, Clang cross-compilation, Docker, TeamCity, GitLab CI, Jenkins, GitHub Actions

Experience

Resideo (Ademco) – CoolPeople contractor

01/2026 – 09/2026

`Test Automation Engineer (Python, C#)`: Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54); sole owner of the verification test suite and of the automated bench that drives it.

- Designed a rigorous `Arrange/Act/Assert` validation pipeline in Python of independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator that expands a `YAML` catalogue of the controller's complete state, event and transition model into a comprehensive executable test suite, replacing hand-written duplication.
- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock: `pyvisa/USBTMC/SCPI` programmable AC source, `CAN` serial-bus monitoring and `Modbus` relay and probe I/O.

- Implemented a correlated `JSONL` timeline placing every instrument command, signal edge and bus message on one axis, enabling root-cause analysis across independent devices.
- Closed a critical coverage gap where stimulus windows were never measured, and fixed a long-hidden leaked USB handle that had been misreported as a hardware fault, unblocking an entire test family.
- Uncovered numerous firmware and harness defects through analysis of the C/C++ sources, each backed by a standalone reproduction script.
- Delivered an unattended nightly bench pipeline (per-key locking, crash and abort recovery, result journalling, Excel evidence workbooks, bilingual HTML dashboards, Mermaid failure graphs) and made report generation dramatically faster.
- Migrated a C#/.NET WinForms serial-bus monitor into version control, fixed a log-parser crash and reduced its build and deployment to a single command.
- Created an agentic AI harness of specialised agents, skills, prompt workflows, policy hooks and `MCP` servers, adopted as a daily part of the delivery workflow.
- Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

  Skills gained: hardware-in-the-loop test design, safety-certified product verification, ignition state-machine domain knowledge, pyvisa/USBTMC/SCPI instrument control, Modbus I/O cards, CAN serial-bus diagnostics, deterministic multi-threaded device ownership, test-evidence and traceability reporting (openpyxl, HTML dashboards, Mermaid), MCP server development, agentic workflow design and governance.

Citadela Resort – freelance client project

08/2026 – 09/2026

`Full-stack Developer (Next.js, TypeScript)`: website and booking back end for a private villa rental business near Brno, live at citadela-resort.cz.

- Built a bilingual (Czech/English) Next.js 15 site whose translation dictionaries are type-checked against each other, with a single-unit booking model and inquiries stored in PostgreSQL through Prisma.
- Integrated Booking.com without a public API: inbound iCal sync, an outbound iCal feed and affiliate deep links, with transactional booking confirmation that prevents double-selling.
- Designed NFC door access and electric-scooter rental: Ed25519-signed reader requests, replay protection, fail-closed access decisions and an audit log with GDPR retention.
- Delivered the brand identity (logo suite, brand book, poster) and deployed on Vercel with Neon Postgres; CI runs type check, unit tests, build and signed-reader checks against the built server.

  citadela-resort.cz | github.com/sparesparrow/citadela

  Skills gained: Next.js 15 and React 19, Prisma 7, Auth.js, Vercel and Neon deployment, iCal integration, Ed25519 device authentication.

Thermo Fisher Scientific – Ixperta contractor

2023 – 2024

`Software Developer (C++)`: Electron-microscopy software for high-resolution detector hardware operation and image acquisition.

- Implemented server-side C++ software with hardware abstraction layers and `COM/MIDL` inter-process communication on Windows.
- Developed embedded software for `FPGA`-based hardware communicating over `gRPC`, in C++ and Python on Linux.
- Contributed to the backend of a C#/WPF image-processing and display application, using COM/MIDL IPC with the HAL and ensuring graceful degradation.
- Built vscode-automation, an enterprise-grade VS Code extension integrating TICS code-quality analysis with the OpenAI API for automated violation detection and AI-powered remediation (TypeScript, VS Code Extension API, Git API).
- Led GitHub Copilot best-practice sessions for the development team and presented bi-weekly stakeholder sprint demos, including the final presentation of the vscode-automation innovation sprint.

  Skills gained: image acquisition and processing, C#/.NET, Component Object Model, Windows development, gRPC, FPGA development, VS Code extension development, TypeScript, OpenAI API integration, technical presentation and stakeholder communication.

Honeywell

2022 – 2023

`Software Engineer II`: Onboard Maintenance System that monitors aircraft condition and diagnoses issues during flight.

- Architected and implemented `DiagTest`, a state-machine diagnostic test engine in C++ managing an extensive set of operational states for aircraft system testing during flight and maintenance.
- Designed job-based asynchronous test orchestration (`DiagTestJob`, `DiagTestMsgQueProc`) with message-queue processing for reliable real-time diagnostics.
- Engineered test-interference management that prevents conflicting diagnostic tests from running simultaneously, protecting aircraft system safety and data integrity.

- Implemented multiple test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.
- Built `MonitorManager`, real-time monitoring of aircraft system parameters during test execution, correlated with maintenance messages.
- Developed comprehensive Python integration tests validating communication protocols, state transitions and stability under failure scenarios.

  Skills gained: embedded software, ARM, RTOS, Clang cross-compilation, FlatBuffers, Conan, Enterprise Architect (UML), aviation safety standards, state-machine design patterns, TeamCity, GitLab CI/CD.

Trusted Network Solutions – Kernun Adaptive Firewall

2020 – 2022

`Application Developer`

- Implemented a C++ HTTP/HTTPS forward proxy on Linux with TLS inspection, OCSP certificate validation, Kerberos authentication and LDAP user/group lookup; CA and client-certificate management, a dynamic configuration builder/parser and a modular manager/partition daemon design.
- Developed DNS resolution and caching, domain categorisation and client authentication methods.
- Integrated Suricata (IDS) and nftables with PostgreSQL log aggregation and connection tracking.
- Developed and maintained `ipmon`, a C++20 network-monitoring daemon: real-time IPv4/IPv6 address tracking over Netlink, dynamic nftables rules, Unix-domain-socket IPC, JSON configuration, RAII and mutex-based thread-safe operations, an event-driven design with configurable delays and atomic configuration writes.
- Rewrote the Perl reporter_log tool in C++ with PostgreSQL integration, structured log parsing and Perl interoperability for legacy compatibility.
- Built a TypeScript configuration generator: YAML-based high-level configuration, certificate-generation automation and integration with the web backend.
- Developed a Perl test framework covering HTTP/HTTPS, TLS inspection, Kerberos/LDAP authentication and OCSP validation.

  Skills gained: TCP and HTTP proxies, encrypted traffic inspection, TLS 1.3, C++20/17, Boost, JsonCpp, Kerberos/Heimdal, OCSP, LDAP, SQLite, Perl, JavaScript/TypeScript, YAML, Jenkins, Linux networking, Netlink, UNIX sockets, SSH tunnelling, port forwarding.

Self-employed – IT services

- B2C: IT support, consumer hardware retail and security consulting.
- B2B: CI/CD automation pipelines and Linux server administration.

  Skills gained: Kali Linux, Wireshark, TeamCity, Docker.

IBM

2014 – 2016

`Automation Developer & IT Systems Specialist`

- Built an automation application that replaced manual workload (ticketing, basic server administration) in Bash, PowerShell and JavaScript, applying automata theory to design state-driven workflows.
- Monitored Unix/Windows production servers and resolved incidents; hands-on with Active Directory and VMware.

  Skills gained: Bash, PowerShell, JavaScript, practical automata theory, Red Hat Linux, Windows Server, Active Directory, VMware.

Selected Open Source

- **MCP-Prompts** — Model Context Protocol server for managing prompt templates and LLM interactions (TypeScript): hexagonal design with memory, file and AWS (DynamoDB, S3, SQS) storage adapters, Docker images per variant and AWS deployment with CDK and Lambda. A popular, actively forked project on GitHub; published on npm and Docker Hub; listed on Glama, MCP Market and AIBASE; Rust reimplementation mcp-prompts-rs, optimised for memory efficiency and high concurrency.
  github.com/sparesparrow/mcp-prompts | github.com/sparesparrow/mcp-prompts-rs
- **MIA** — vehicle telemetry and IoT platform across Android (Kotlin), Raspberry Pi (Python, FastAPI, ZeroMQ), C++ and ESP32. OBD-II digital twin, hardware simulation fallback, read-only vehicle safety profile, MCP modules.
  github.com/sparesparrow/mia | case study
- **mcp-project-orchestrator** — JSON-driven orchestration that recognises design patterns in a request and scaffolds project files from templates (variable substitution, prompt-template rendering), with Mermaid flowchart, sequence and class diagrams rendered to SVG/PNG (Python). Listed on PulseMCP; featured in a Skywork deep dive.
  github.com/sparesparrow/mcp-project-orchestrator
- **hard-coder** — early agentic reasoning and planning assistant that turns ideas into working code with tests and documentation, designed before LLMs had built-in reasoning; evolved from the OpenAI GPT Store through Anthropic and CrewAI to Cursor Composer.
  github.com/sparesparrow/hard-coder

- **github-events** — Python service monitoring a broad range of GitHub event types: repository health scoring, developer-productivity analysis and security anomaly detection; full FastAPI REST API, live dashboard, MCP server support, SQLite or AWS DynamoDB backends.
  github.com/sparesparrow/github-events
- **human-action** — PDF-to-audiobook pipeline (text clean-up and chunking, speech synthesis with Piper TTS or the ElevenLabs API, chapter assembly); produced the Czech audiobook of Mises' Human Action on YouTube.
  github.com/sparesparrow/human-action
- **rtp-midi** — modular Rust architecture for low-latency MIDI, audio and LED control: RTP-MIDI to DAWs and OSC to addressable LEDs; Android hub (Kotlin + Rust NDK, foreground service, AMidi, mDNS discovery) and an ESP32 visualiser (FastLED, dual-core FreeRTOS); Docker-based CI/CD.
  github.com/sparesparrow/rtp-midi
- **ipmon** — C++20 Netlink daemon that detects IPv4/IPv6 address changes and updates nftables rules; RAII, move semantics, thread-safe atomics, Unix-socket IPC, JSON configuration, Debian packaging with systemd hardening.
  github.com/sparesparrow/ipmon
- **rust-network-mgr** — Rust redesign of ipmon: atomic nftables updates via Netlink that prevent inconsistent rulesets during topology changes, lock-free structures and epoll-based event processing; local HTTP API and an MCP server for AI tools. Published and actively downloaded on crates.io.
  github.com/sparesparrow/rust-network-mgr
- **sparetools** — monorepo of shared Python tools and Conan recipes with a bundled CPython runtime (cpy), demonstrating C/C++ dependency management and cross-platform CI/CD.
  github.com/sparesparrow/sparetools
- **podman-desktop-extension-mcp** — Podman Desktop extension that combines agent containers and routes collaboration between them.
  github.com/sparesparrow/podman-desktop-extension-mcp

Education & Training

- **Masaryk University, Faculty of Informatics** — Computer Science coursework *(no degree awarded)*: programming, Linux, networking.
- **DeepLearning.AI / Coursera** — Deep Learning Specialization (Andrew Ng), data analysis, prompt engineering, building LLM agents.
- **Self-directed** — Hack The Box; design patterns; Certified Ethical Hacker syllabus *(exam not taken)*; *Software Engineering at Google*, *The Pragmatic Programmer*, *Clean Architecture*.

References available on request.